**VIJAY VENKATA SESHA SHAIE THOTA**

**Senior Forward Deployed AI Engineer | Generative AI & Agentic AI Engineer**

**LLMs • RAG • Multi-Agent & Multi-Model Orchestration • LangChain • LlamaIndex • MCP • AI Evaluation • Enterprise AI Integration**

**vijaythota2509@gmail.com|Austin, Texas|+1 (669)-388-1181|Availability: Immediate|Open to customer-site travel up to 50%|**

## Professional Summary :

Senior Forward Deployed AI Engineer and AI Solutions Architect offering over 9 years of enterprise experience designing, deploying, and optimizing production-grade Conversational AI, Agentic workflows, and cloud-native systems. Proven track record leading customer-facing delivery from technical discovery workshops and Customer User Journey (CUJ) mapping through rapid prototyping, production rollout, and post-deployment optimization. Possesses deep technical specialization across Python platform engineering, FastAPI, Gemini, Google ADK, Vertex AI Agent Builder, Vertex AI Agent Engine, Customer Engagement Suite (CES), Contact Center AI (CCAI), LangGraph orchestration, MCP protocols, ReAct reasoning, Enterprise RAG, Terraform infrastructure, automated evaluation pipelines, and distributed observability. Expert in integrating autonomous agents with secure enterprise APIs, legacy backends, vector knowledge bases, and identity frameworks while driving root-cause analysis for complex tool execution, retrieval precision, and operational latency.

## Core Competencies :

- **Python Platform Engineering:** Python, asyncio, async/await, FastAPI, Flask, Pydantic, type hints, abstract base classes, OOP, REST APIs, GraphQL, gRPC, microservices, exception handling, asynchronous processing, scalable services.
- **Claude & MCP Agent Engineering:** Anthropic Claude Messages API, Claude Agent SDK, Tool-Using Agents, MCP Servers, Streamable HTTP, Authenticated MCP Gateways, Structured Tool Responses, Prompt Caching, Static Prompt Prefixes, Session Management, Server-Side Tool Enforcement, Failure Recovery, Production Agent Integration.
- **Forward Deployed Engineering:** Customer Technical Discovery, Customer User Journeys (CUJs), Conversational AI Pilots, Rapid Prototyping, Customer Co-Building, Production Deployment, Post-Deployment Optimization, Production Troubleshooting, Technical Workshops, Partner Engineering Enablement, Product Feedback Loops, Reusable Solution Accelerators
- **Agentic AI & Multi-Agent Systems:** Google ADK, LangGraph, ReAct, Self-Reflection, Multi-Agent Orchestration, MCP, A2A, Agent Memory, Session Management, State Management, Tool Calling, Dynamic Tool Selection, Failure Recovery, Human-in-the-Loop.
- **Google Cloud & Conversational AI:** Gemini, Google ADK, Vertex AI Agent Builder, Vertex AI Agent Engine, Dialogflow CX, CX Agent Studio, Agent Assist, Vertex AI Search, Vertex AI Vector Search, Cloud Run, GKE, BigQuery, Cloud Logging, Cloud Trace, Cloud Monitoring.
- **Enterprise RAG & Knowledge Grounding:** Enterprise RAG, Vertex AI Search, Azure AI Search, OpenSearch, Pinecone, Vector Search, Hybrid Retrieval, Embeddings, Document Chunking, Cross-Encoder Reranking, Context Grounding, Knowledge Bases, Hallucination Detection, Retrieval Evaluation.
- **Cloud Infrastructure & DevOps:** GCP, Azure, AWS, Terraform, Cloud Run, GKE, AKS, Kubernetes, Docker, Serverless Functions, IAM, Networking, Secrets Management, GitHub Actions, Cloud Build, CI/CD.
- **Full-Stack & Enterprise Integration:** Python, FastAPI, TypeScript, Node.js, React.js, Next.js, REST APIs, GraphQL, gRPC, Microservices, OAuth2, OIDC, JWT, RBAC, Legacy Systems, Enterprise APIs, Event-Driven Architecture

## Education :

- **Master of Science in Information Systems / Studies —** Trine University, Detroit, MI | Graduation Year: Dec 2023
- **Bachelor of Science in Computer Science / Applications —** Aurora Degree College, Hyderabad, India | Graduation Year: March 2017

## Certifications :

- AWS Certified Generative AI Developer – Professional
- Google AI Essentials (Coursera / Google)
- Generative AI: Concepts, Techniques, and Applications (Fractal)

## Technical Skills :

| Category | Deep Technical Specialization |
|---|---|
| **Agentic AI & Multi-Model Orchestration** | LangGraph, CrewAI, Google ADK, MCP, A2A, ReAct, Self-Reflection, Multi-Agent Orchestration, Multi-Model Orchestration, Dynamic Model Routing, Model Selection, Fallback Strategies, Tool Calling, Agent Memory, Session Management, Human-in-the-Loop, structured tool responses, fail-open/fail-closed. |
| **Generative AI** | Gemini, Claude, Claude Messages API, Claude Agent SDK, OpenAI GPT-4/5, Azure OpenAI, Azure AI Foundry, Prompt Engineering, Function Calling, Tool Calling, Prompt Registry, Prompt Versioning, Model Registry, Model Versioning, Semantic Caching, Cost/Latency Optimization, prompt caching, static prompt prefixes. |
| **Prompt Engineering** | System and task prompt design, few-shot prompting, structured outputs, tool-use prompting, prompt versioning, prompt regression testing, grounding optimization, context engineering, and production prompt evaluation. |
| **Google Conversational AI & CX** | Gemini, Google ADK, Vertex AI Agent Builder, Vertex AI Agent Engine, Conversational Agents (Gemini-powered CX), Customer Engagement Suite (CES), Contact Center AI (CCAI), Dialogflow CX, CX Agent Studio, Agent Assist, CCaaS, Voice AI, Voice Agents, SCRAPI Integration Patterns, Customer User Journeys (CUJs) |
| **Enterprise RAG** | Hybrid Search, Vertex AI Search, Knowledge Graph, Enterprise Grounding, Grounding APIs, Hybrid Retrieval, Context Engineering, RAG Technologies |
| **Vector Databases** | Azure AI Search, Pinecone, ChromaDB, FAISS, OpenSearch, Vector Search, Vertex AI Vector Search |
| **Enterprise Integrations** | REST APIs, GraphQL, gRPC, Microservices, Event-Driven Architecture, API Integration, Asynchronous Processing, Scalable Services, MCP Streamable HTTP, authenticated MCP gateways |
| **Agile & Engineering Collaboration** | JIRA, Confluence, Agile/Scrum, sprint planning, technical documentation, architecture reviews, design reviews, and cross-functional delivery. |
| **Backend** | FastAPI, Flask, Express.js, Backend Frameworks, asyncio, async/await |
| **Frontend** | React.js, Next.js, HTML5, CSS3, Tailwind CSS |
| **Salesforce Agentforce** | Working knowledge of Agentforce architecture, agent topics, instructions, actions, knowledge grounding, tool/API integration, and enterprise agent orchestration patterns |
| **Cloud** | Azure AI Foundry, Azure AI Services, Azure AI Search, Azure Functions, Azure Kubernetes Service (AKS), Azure DevOps, Google Cloud Platforms. |
| **Evaluation & Observability** | RAGAS, TruLens, LangSmith, OpenTelemetry, Arize Phoenix, Cloud Logging, Cloud Trace, Cloud Monitoring, Regression |

| | |
|---|---|
| | Evaluation, Grounding Evaluation, Hallucination Detection, Reasoning Evaluation |
| **Google Conversational AI & CX** | Conversational Agents (Gemini-powered CX), Customer Engagement Suite (CES), Contact Center AI (CCAI), Dialogflow CX, CX Agent Studio, Agent Assist, CCaaS Integration Patterns, Voice AI, Voice Agents, Customer User Journeys, Gemini, Google ADK, Vertex AI Agent Builder, Agent Engine, Multi-Turn Conversation Management, Human Escalation, Enterprise API Integration, SCRAPI Integration Patterns. |
| **DevOps** | Terraform, GitHub Actions, GitLab CI/CD, Azure DevOps, Cloud Build, Vertex AI Pipelines, Google Cloud Deploy, Docker, Kubernetes, AKS, GKE, AI Reliability Engineering |
| **Observability** | Google Cloud Logging, Cloud Trace, and Cloud Monitoring instrumentation, Arize Phoenix, LangSmith, OpenTelemetry, Centralized Logging, Distributed Tracing, Prometheus, Grafana, SLIs, SLOs, Error Budgets, Incident Response, AI Governance, Responsible AI |
| **Evaluation** | Azure AI Foundry Evaluation, Prompt Flow Evaluation, RAGAS, TruLens, Prompt Regression Testing, Model Regression Testing, Continuous Validation Pipelines, Automated Evaluation Pipelines, Grounding Evaluation, Reasoning Evaluation, Hallucination Detection. |
| **Security & Agent Identity** | OAuth2/OIDC, OAuth client credentials, JWT validation, JWT/JWKS, RBAC, secrets management, authenticated MCP access, server-side authorization, identity-bound sessions, PII-safe logging, policy enforcement, HIPAA, SOC 2, PCI-DSS. |
| **Data Engineering** | Databricks, Airflow, Spark SQL, Delta Lake, ETL/ELT, BigQuery |
| **Programming Languages** | Python, Python type hints, ABC/subclassing, TypeScript, Node.js, SQL, PySpark, Bash |
| **Experimentation** | A/B Testing, Personalization, Real-Time Decisioning, Optimization Logic |
| **Enterprise Integration & Security** | Gemini API, Azure OpenAI API, OpenAI APIs, Claude/Anthropic APIs, function calling, structured outputs, tool calling, REST integration, authentication, rate limiting, retries, and production error handling. |

## CORE PLATFORM ENGINEERING EXPERTISE

- **AI-Assisted Development:** GitHub Copilot, AI Pair Programming, AI-Assisted Coding, Code Refactoring, Test Generation, Code Review, Prompt-Based Development, Responsible AI-Generated Code Validation
- **API Engineering:** OpenAPI, Swagger, Contract-First Design, API Versioning, Rate Limiting, OAuth2, OIDC, JWT, Authentication, Authorization
- **Python Quality Tooling:** Pytest, Ruff, Black, mypy, Static Type Checking, Code Coverage, Unit Testing, Integration Testing, Contract Testing, Load Testing
- **SDK Engineering:** Python SDKs, PyPI, Python Packaging, pyproject.toml, Semantic Versioning, Developer Documentation, Code Samples, Backward Compatibility

## Professional Experience:

**CLIENT: CHASE — Remote** **Austin, Texas**

***Senior Forward Deployed AI Engineer | AI Solutions Architect*** *June 2024 – Present*

**Drives end-to-end Forward Deployed AI engagements** from initial technical discovery and **Customer User Journey (CUJ) mapping** through rapid prototyping, agentic system design, full-stack microservice development, automated evaluation, and **production-grade deployment**, accelerates enterprise platform

adoption by turning strategic AI use cases into highly reliable, scalable solutions with clear operational ROI.

- **Collaborated with cross-functional technical stakeholders** to convert intricate **Customer User Journeys (CUJs)** into robust **Conversational AI** architectures; mitigated operational friction by streamlining **state management**, identity controls, enterprise network integration, and cloud infrastructure requirements.
- **Co-built AI solutions** with **customer engineering teams** through technical discovery, **architecture workshops**, hands-on development, **design reviews**, **troubleshooting**, and **production-readiness validation**, establishing reusable engineering practices that supported long-term platform adoption.
- **Architected and productionized Agentic AI systems** on GCP leveraging **Gemini**, **Google ADK**, **Vertex AI Agent Builder**, and **Vertex AI Agent Engine**; orchestrated containerized deployments across Cloud Run and GKE while integrating secure enterprise APIs to convert conversational pilots into **production-grade delivery**.
- **Engineered Python-based tool and MCP integration layers** for autonomous LLM systems utilizing **asyncio paradigms**, typed Pydantic interfaces, modular backend microservices, **structured tool responses**, server-side authorization protocols, and **deterministic failure recovery** across low-latency **enterprise APIs**.
- **Engineered identity-bound agent security frameworks** incorporating **OAuth2/OIDC protocols**, JWT validation, JWKS token verification, OAuth client credentials, secrets management, and **policy-driven tool authorization**; established risk-calibrated fail-safe mechanisms alongside **PII-compliant audit logging** across high-consequence enterprise interactions.
- **Productionized Anthropic Claude integrations** leveraging the **Messages API** and **Claude Agent SDK** to construct autonomous tool-calling workflows; optimized performance via **prompt caching** and static instruction prefixes while enforcing **structured tool responses**, robust retry strategies, timeout management, identity-bound session state, and automated **regression testing**.
- **Architected secure MCP server integrations** utilizing **Streamable HTTP** to expose core platform capability via **authenticated gateways**; enforced **server-side authorization**, cryptographically signed identity assertions, scoped API permissions, **structured tool responses**, and policy-driven controls to safeguard high-consequence backend functionality.
- **Engineered real-time trust and risk decisioning mechanisms** within autonomous agent execution paths; converted enterprise backend signals into standardized approval, rejection, authentication step-up, service unavailability, and **structured tool responses** while maintaining operation-specific fail-open and fail-closed security posture under ultra-low **latency constraints**.
- Developed reusable integration patterns for customer-facing transactional agent workflows, separating session identity, tool authorization, decisioning, backend execution, outcome reporting, and audit controls to reduce implementation effort for downstream engineering teams.
- **Architected Conversational AI and CX systems** leveraging **Dialogflow CX**, **CX Agent Studio**, **Agent Assist**, **Gemini**, and **Google ADK**; connected multi-turn interactive agents with secure **enterprise APIs**, identity platforms, vector repositories, legacy backends, and **human escalation workflows**.
- **Architected multi-agent workflows** using **LangGraph**, **Google ADK, ReAct reasoning**, **self-reflection**, **MCP**, **persistent state**, and **Human-in-the-Loop controls**, enabling dynamic tool selection, **failure recovery**, and controlled autonomous execution.
- **Engineered secure API and identity integration patterns** bridging autonomous AI agents with enterprise infrastructure utilizing **FastAPI microservices**, REST endpoints, **MCP tool protocols**, vector search engines, legacy backends, OAuth2/OIDC, JWT authentication, RBAC, and policy-driven security controls.
- **Architected declarative Terraform infrastructure** blueprints to automate cloud provisioning for agentic microservices, serverless workloads, fine-grained IAM controls, virtual networking, secrets management, and containerized Kubernetes clusters across dev, test, and production environments.
- **Instrumented full-stack AI services using OpenTelemetry**, Google Cloud Logging, and Cloud Trace; mapped end-to-end conversation identifiers across backend microservices to perform real-time root-cause analysis on **Agentic AI reasoning** flaws, invalid tool invocation, integration exceptions, vector retrieval defects, and execution latency overhead.

- **Constructed automated evaluation and validation pipelines** across 300+ enterprise scenarios targeting **prompt regression testing**, ReAct step execution, dynamic tool selection, **context grounding**, **hallucination detection**, policy guardrails, and operational latency; **curtailed production deployment failures by 45%** while facilitating continuous platform refinement.
- **Architected production-scale Enterprise RAG** frameworks incorporating Azure AI Search, OpenSearch, and Pinecone; refined semantic chunking, vector embeddings, **hybrid retrieval** pipelines, cross-encoder reranking, context grounding, and automated **hallucination detection** to attain 92% retrieval precision.
- **Executed real-time root-cause analysis** for high-throughput intelligent platforms during mission-critical operational spikes; resolved complex bottlenecks across **Agentic AI reasoning** flows, dynamic tool invocation, **secure enterprise APIs**, vector knowledge bases, infrastructure routing, and **IAM security controls** to maximize uptime and platform stability.
- **Designed multi-model LLM orchestration and inference optimization** spanning Gemini, Claude, and Azure OpenAI platforms; integrated **dynamic model routing**, automated fallback strategies, **semantic caching**, prompt tuning, and token governance to **curtail operational expenses by 42%** while boosting system throughput by 29%.
- **Analyzed systemic deployment bottlenecks and technical roadblocks** across client landscapes; synthesized these operational insights to construct **reusable solution accelerators** including modular Python libraries, **MCP tool protocols**, declarative **Terraform infrastructure** blueprints, automated validation suites, and enterprise deployment frameworks, while establishing direct **product feedback loops** to drive core platform enhancements.
- **Co-built AI solutions** with **customer engineering teams** through technical discovery, **architecture workshops**, hands-on development, **design reviews**, **troubleshooting**, and **production-readiness validation**, establishing reusable engineering practices that supported long-term platform adoption.

***Environment:*** *Python, FastAPI, TypeScript, React.js, Gemini, Google ADK, Vertex AI Agent Builder, Vertex AI Agent Engine, Dialogflow CX, Vertex AI Search, Cloud Run, GKE, Google Cloud Logging, Cloud Trace, Azure AI Foundry, Azure OpenAI, LangGraph, LangChain, MCP, ReAct, OpenSearch, Pinecone, RAGAS, TruLens, OpenTelemetry, Terraform, GitHub Actions, Docker, Kubernetes, OAuth2, OIDC, JWT, RBAC.*

**CLIENT: PACIFIC GAS AND ELECTRIC COMPANY** **San Francisco, CA**
***AI Solutions Engineer | Intelligent Automation Architect*** *June 2022 – May 2024*

**Architected and deployed production-scale Enterprise AI** and intelligent automation architectures empowering critical utility ecosystems; integrated **real-time streaming telemetry**, predictive ML pipelines, **secure enterprise APIs**, cloud-native infrastructure, and comprehensive **operational observability** to ensure continuous reliability.

- **Architected and operated production-scale Python and FastAPI services** integrating Azure OpenAI, LangChain, Azure AI Search, and AKS to transform ambiguous operational requirements into scalable, AI-assisted workflows.
- **Collaborated with business stakeholders and field operations** to drive **rapid prototyping** and requirement validation, accelerating **production-grade adoption** across complex enterprise and operational workflows.
- **Executed post-deployment solution optimization** by leveraging operational insights and customer feedback loops to maximize **retrieval quality**, platform adoption, and the overall **effectiveness of Enterprise AI** across production environments.
- **Designed secure REST APIs with structured validation** for GIS, telemetry, and enterprise platform integrations, ensuring robust connectivity for predictive modeling systems.
- **Built enterprise RAG and semantic-search services** over maintenance runbooks and incident reports using Azure AI Search, ChromaDB, and vector search patterns to improve retrieval accuracy and grounding.
- **Reduced field-technician mean time to resolution (MTTR) by 65%** by implementing grounded access to decades of historical maintenance and incident knowledge.
- **Containerized and deployed scalable services on AKS** using Docker, Terraform, and GitLab CI/CD, while standardizing reusable cloud-infrastructure templates and deployment blueprints.

- **Reduced deployment time to under ten minutes** through automated testing, infrastructure provisioning, and streamlined deployment pipelines.
- **Implemented automated data-quality and performance monitoring** using Airflow and PySpark, establishing continuous validation pipelines to detect model drift and degradation.
- **Instrumented production services with comprehensive logging, metrics, and monitoring** to identify performance bottlenecks and maintain platform reliability.
- **Processed over 10 million daily telemetry events with sub-minute latency** using distributed PySpark and Azure Event Hubs pipelines.
- **Partnered with cross-functional stakeholders** to define requirements, deliver practical MVPs, communicate architectural tradeoffs, and drive service adoption.

***Environment:*** *Python, FastAPI, Azure OpenAI, Azure AI Search, AKS, Docker, Kubernetes, Terraform, GitLab CI/CD, LangChain, ChromaDB, REST APIs, Pydantic, Prometheus, Airflow, PySpark, SQL, Azure Event Hubs.*

**CLIENT: FORSYS INC** **Hyderabad, India**
***Software Engineer | AI Platform Engineer*** *April 2017 – July 2021*

- **Engineered scalable Python and FastAPI REST and GraphQL services** that securely exposed model predictions and enterprise data with sub-50ms p95 response latency.
- **Built cloud-native microservices and event-driven workflows** using Python, Node.js, Docker, Azure Synapse, Azure Event Hubs, PostgreSQL, and Airflow.
- **Implemented secure authentication and authorization** using OAuth2, JWT, RBAC, Azure Key Vault, and secrets-management practices.
- **Established Azure DevOps CI/CD pipelines** with pytest, Mocha, and Jest unit and integration testing, automated quality gates, and more than 90% code coverage.
- **Developed automated data-quality assertions and validation pipelines** that prevented invalid or incomplete data from reaching downstream models and services.
- **Created centralized operational dashboards** for pipeline health, data quality, service latency, model drift, and platform reliability.
- **Applied HIPAA and SOC 2 security requirements** across healthcare data services, APIs, access controls, auditability, and protected workflows.
- **Collaborated with technical and business stakeholders** and developed reusable deployment patterns that improved delivery consistency across engineering teams.

***Environment:*** *Python, FastAPI, Flask, REST APIs, GraphQL, Pytest, Azure DevOps, Docker, Azure Synapse, Azure Data Factory, PostgreSQL, SQL, PySpark, Airflow, Azure Event Hubs, OAuth2, JWT, RBAC, Azure Key Vault, HIPAA, SOC 2*

**SocialWave – Emotionally Intelligent AI Platform (Startup / Founder)**
***Founder | AI System Architect***

Architected and engineered SocialWave, an enterprise-grade emotionally intelligent AI platform harmonizing LLMs, Agentic Orchestration, and production-scale RAG pipelines to facilitate adaptive AI experiences. Demonstrated mastery in integrating multimodal intelligence, situational reasoning, and Human-in-the-Loop (HITL) gates within a cloud-native ecosystem. Proven track record in delivering high-throughput workflow automation, vector-based memory, and secure microservices that optimize performance and ROI across mission-critical digital wellness and customer engagement sectors.

**Key Objectives & Vision**

Architected a consolidated emotional intelligence framework to bridge human-AI interactions, optimizing interpersonal communication, clinical wellness assistance, and enterprise-grade customer engagement through the integration of multimodal empathy, real-time situational awareness, and complex reasoning within a unified production AI ecosystem.

**Impact / Outcomes:**

- Architected a sophisticated multi-tier emotional AI framework, harmonizing real-time sentiment detection, situational intelligence, and LLM-driven cognitive reasoning.

- Engineered an enterprise-scale system blueprint for high-throughput Emotion AI, optimized for personal companions, corporate customer experience, and clinical wellness solutions.
- Established the core orchestration layer for empathetic AI agents, integrating conversation memory, hyper-personalization, and synchronous emotional data signals.
- Developed a strategic product roadmap spanning B2C wellness applications, B2B emotional intelligence SaaS, and a developer-centric API ecosystem.
- Positioned the platform as a next-generation emotional intelligence abstraction layer, enhancing the capabilities of conventional LLM-based conversational assistants.

**Key Contributions:**

- Architected a comprehensive Emotion AI system architecture integrating multimodal inputs, real-time sentiment detection, cognitive reasoning, and persistent memory orchestration layers.
- Designed multimodal conversational AI pipelines to process and synchronize textual, vocal, and facial signals for empathetic response generation, establishing voice-aware interaction patterns, session context, and secure API-driven orchestration applicable to enterprise Voice AI and Voice Agent experiences.
- Developed adaptive reasoning frameworks that fuse LLMs with real-time context APIs including weather and scheduling data enabling situationally aware and personalized user interactions.
- Built persistent emotional memory architectures using vector databases to enable longitudinal mood tracking and hyper-personalized context across multi-turn human-AI conversations.
- Defined agentic AI workflows and orchestration patterns specifically tailored for digital companionship, enterprise customer experience, and clinical wellness applications.
- Engineered a privacy-first system design incorporating automated data anonymization, end-to-end encryption, and local-processing protocols to ensure high-security emotional data handling.
- Established a strategic platform roadmap, defining the MVP scope, scalable monetization models, and a comprehensive SDK/API ecosystem for enterprise-grade Emotion AI adoption.
- Engineered a Google ADK-based multi-agent orchestration layer leveraging Vertex AI Agent Builder, Agent Engine, and the Gemini API, incorporating agent memory, session management, and A2A/MCP collaboration protocols for autonomous companion and enterprise agents.
- Extended the Google Cloud agent architecture toward enterprise conversational and Voice AI use cases, designing integration patterns for multi-turn customer journeys, voice interactions, human-agent escalation, authenticated enterprise APIs, and contact-center workflows aligned with Dialogflow CX, CX Agent Studio, Agent Assist, and CCaaS architectural patterns.
- Designed secure API and identity integration patterns for conversational agents across enterprise services, incorporating user authentication, OAuth2/OIDC, policy-controlled tool access, session context, and reusable service interfaces applicable to telecom and enterprise business architectures.
- Implemented enterprise-grade retrieval and knowledge grounding using Vertex AI Search, Vertex AI Vector Search, and Knowledge Graph structures, deployed on Cloud Run and GKE with BigQuery-backed analytics, reinforced by AI Governance and Responsible AI safeguards.

***Environment:*** *Python, FastAPI, REST APIs, Microservices, AWS Lambda, Amazon ECS, Docker, Kubernetes, Terraform, OAuth2, JWT, OpenTelemetry, Azure OpenAI, Amazon Bedrock, LangChain, LangGraph, LlamaIndex, RAG, Prompt Engineering, LLM Evaluation, Guardrails, Pinecone, ChromaDB, FAISS, AI Agents, Multi-Agent Orchestration, HITL, RAGAS, MCP, AI Governance.*